B.Tech PROJECT

OPTIMAL CONTROL OF MULTI-AREA POWER SYSTEM

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We, Aryan Prakhar (2021UEE4610), Chirag Maini (2021UEE4659), Deepanshu (2021UEE4617) and Pallavi Dheeman (2021UEE4583) students of B.Tech., Department of Electrical Engineering, hereby declare that the Project-Thesis titled "***OPTIMAL CONTROL OF MULTI-AREA POWER SYSTEM***" which is submitted by us to the Department of Electrical Engineering, Netaji Subhas University of Technology (NSUT), Dwarka, New Delhi in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology is our original work and not copied from any source without proper citation. The manuscript has been subjected to plagiarism checks by Turnitin software. This work has not previously formed the basis for the award of any other degree.

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CERTIFICATE OF DECLARATION

DEPARTMENT OF ELECTRICAL ENGINEERING

This is to certify that the work embodied in project thesis titles, "**OPTIMAL CONTROL OF MULTI-AREA POWER SYSTEM"** by Aryan Prakhar (2021UEE4610), Chirag Maini (2021UEE4659), Deepanshu (2021UEE4617) and Pallavi Dheeman (2021UEE4583) is the bonafide work of group submitted to **Netaji Subhas University of Technology** for consideration in 8th semester B.Tech. Project Evaluation.

The original Research work was carried out by the team under my/our guidance and supervision in the academic year 2024-2025. This work has not been submitted for any other diploma or degree of any university. On the basis of declaration made by the group, we recommend the project report for evaluate

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ABSTRACT

This thesis presents the Optimal Control ways for a multi-area power system to enhance frequency regulation and inter-area stability under varying load conditions. This study aims to model multi-area multi-source systems with the presence of units such as thermal, hydro, diesel, and gas units, wind farms, EVs, and energy storage sources. Without a properly coordinated controller, they can expose the system to severe stress under unwanted disturbances and cause it to deviate from normal operating conditions. Therefore, after modelling the proposed system, a Proportional–Integral–Derivative (PID) controller based on a low-pass filter is presented as a widely used device in the industry.

Tuning coefficients of this controller based on trial and error were irrational, this issue turned to a frequency-based optimization problem with different operating points in unwanted operating conditions in order to have a robust design. In the previous articles, a fixed working point was employed to design the controller. To cover this issue, modelling towards reality, and a wider range of unexpected conditions and events, ±30% design uncertainty and different loading conditions are considered in this study.

Different scenarios based on unwanted disturbances, uncertainties, and random load changes are considered to evaluate the system and algorithm performance. Numerical results obtained by analysing comparative criteria on time and frequency domains indicated the proposed method had over 25% better performance compared to other methods, on average. The results obtained from the frequency and time domains properly show that the settlement and overshoot and undershoot times of the system equipped with the proposed controller are less compared to other designs in the literature, and better stability has been provided in the s-plane

Keywords: Optimal Control, Multi-Area Power System, Frequency Regulation, PID Controller, Frequency Based Optimization.

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Chapter 1

Introduction

1.1 Motivation

The growing complexity and the integration of varied energy sources into modern power grids motivate this project. Hydro, diesel, gas, wind, electric vehicles as well as storage systems are examples of such sources. Multi-area systems face changing and often unpredictable load conditions. Frequency stability plus inter-area coordination faces increased difficulty under those conditions. Traditional control methods based on fixed operating points do not handle such dynamic situations well. A clear need for tough and adaptable control strategies appears. Optimal PID control with frequency-based optimization is one example. Power system operation needs to be reliable and stable, even with disturbances, uncertainties next to much system change.

1.2 Key Challenges

One of the biggest challenges in this project is handling the complexity of a modern power system that brings together many different energy sources—like thermal, hydro, diesel, gas, wind, EVs, and batteries. Each of these behaves differently, so getting them to work smoothly together is complex. On top of that, power demand constantly changes with varying load demand in any area or region, and unexpected disturbances can throw the system off balance. Making sure the system stays stable and maintains proper frequency under all these conditions is an important task for maintaining the grid system and completing the demand.

Another challenge is tuning the PID controller effectively. Doing it by trial and error doesn’t work well, especially with so many variables at play. That’s why we’ve turned to frequency-based optimization methods, which help the controller adapt to different conditions and uncertainties—like changes in system behaviour of up to ±30%. Building this kind of smart, flexible control system and testing it thoroughly through simulations

takes time and careful design, but it's essential for making future power systems more reliable and efficient.

1.3 Problem Addressed in Thesis

This thesis focuses on real-world problems faced by today's complex power system. Now there is availability of more diverse energy sources like thermal, hydro, diesel, gas, wind, EVs, and batteries are added to the grid, keeping everything stable and system's frequency become harder, especially when load demand constantly changes or unexpected disturbances occur. PID controllers are usually tuned using trial and error, which doesn't work well when the system is unpredictable or constantly shifting.

1.4 Approach to the Problem and Organization of Thesis

The approach taken in this thesis starts by building a realistic model of a multi-area power system that includes different energy sources like thermal, hydro, diesel, gas, wind farms, EVs, and energy storage. Since these sources behave differently and interact in complex ways, managing stability—especially frequency—is a major challenge. To address this, a PID controller is chosen due to its simplicity and widespread use. However, instead of relying on traditional trial-and-error tuning, the controller is optimized using a frequency-based method to ensure it performs well even under unpredictable load changes and up to ±30% design uncertainty. The system's response is tested through various disturbance scenarios, and its performance is evaluated using detailed simulations in MATLAB/Simulink, focusing on both time and frequency domain behaviour.

The thesis is organized in a way that walks the reader through the entire journey of the project. It begins with an introduction that explains the background, motivation, and goals of the study. This is followed by a literature review, which looks at existing methods and highlights where they fall short. The next chapter dives into how the power

system is modelled, covering all the energy sources included. Then, the controller design and the optimization process are explained in detail. After that, simulation results are presented along with a comparison of how the system performs under different conditions. Finally, the thesis wraps up with conclusions and suggestions for future work, offering insights into how this research can be built upon.

Chapter 2

Literature Review

2.1 Literature Survey

Table 2.1 presents a summary of various approaches employed by different researchers for optimisation of the Multi Area System. The table includes authors' names, publication years, methods utilized, and the average prediction error.

2.2 Review of Estimation Methods

Various documents classify techniques for optimal control of load in multi area systems, estimation differently. Some categorize them as experiment-based, model-based, or machine-learning-based, methods, while others classify them into direct, indirect, adaptive, and data-driven methods. In the literature, several studies and proposals of LFC (load frequency control) and their design processes have been introduced The robust control theory, the internal model-based controllers, the decentralized control schemes, the sliding mode controllers, the adaptive control systems, the intelligent control systems, and the model predictive controllers have found wide applications in the literature, Additionally, the classical integer order control methods have shown good performance with disturbance rejection capabilities. Furthermore, the fuzzy logic-based control methods have been integrated with various control methods to adapt and enhance the performance of the LFC at various operating conditions and disturbances. In addition, recent deep learning and data-driven LFC schemes have been presented to solve the frequency regulation problems. However, they need huge training data to optimize their performance.

From another point of view, the classical PID controller, comprising proportional, integral, and derivative components, is a widely used method for Load Frequency Control (LFC) in power systems. In this context, the integral and derivative actions play crucial roles in maintaining frequency stability. The integral component

eliminates steady-state errors by accumulating frequency deviations over time and adjusting the control input accordingly. This ensures that the system frequency returns to its nominal value after load disturbances. The derivative component, on the other hand, predicts the future trend of the frequency deviation by considering its rate of change, which improves the transient response and helps in damping oscillations. Together, these two actions enhance system stability, reduce frequency fluctuations, and support optimal control without the need for advanced techniques.

Although simple, this classical approach remains effective for managing the dynamic behaviour of multi-area power systems and serves as a solid foundation for further control system enhancements.

While many modern approaches utilize advanced tools, optimization algorithms, several scientists have explored traditional estimation techniques that rely on analytical and empirical observations of system behaviour. One such approach, discussed and summarized, outlines a range of classical estimation methods that have been effectively used in the past for PID-based control in power systems.

Estimation methods play a critical role in the monitoring and control of multi-area power systems, particularly for load frequency control (LFC) and state estimation. Traditional model-based estimation techniques, such as the Kalman Filter (KF) and its variants—Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF)—have been widely employed to estimate system states and disturbances with high accuracy under linear and mildly nonlinear conditions. These methods rely on accurate system models and noise statistics, enabling real-time tracking of frequency deviations and tie-line power flows. The tilt-integral-derivative (TID)-based LFC has been widely introduced in the literature.

The TID LFC methods possess improved feedback due to their transfer function, which is very close to the optimum. In, a TID-based LFC was presented with a high-pass filter to reduce existing input noise effects on the system stability. Another application of TID LFC was presented in, and it was optimized using the artificial bee-colony optimizer

algorithm (ABC). Another LFC method based on the TID control method was introduced in. The optimization process was made using the pathfinder optimization algorithm (PFA). The differential evolution optimizer algorithm (DE) was introduced in for optimizing the TID control with a filter for LFC methods. A combined PI and TD with a filter controller was proposed in with the slap swarm optimization algorithm (SSA). A parallel combination of TID with a filter and TID was proposed in, and a hybrid algorithm of a modified particle swarm optimizer with the genetic algorithm (MPSOGA) was presented for designing the controller.

Additionally, combined and hybrid FO-based LFC methods have been presented in the literature. A hybrid FO controller was proposed in for gate-controlled series-capacitor (GCSC)-based interconnected power systems. Another hybrid FO LFC was proposed in for superconducting energy storage system (SMES)-based interconnected two-area power systems. A parallel combination of the hybrid FO LFC with the TID with a filter was presented in with an optimum design of the presented controller. A modified hybrid decentralized FO-based LFC and EV controller was presented in. The presented LFC and EV controllers combine the FOPID and TID controllers. Moreover, the artificial ecosystem optimizer (AEO) algorithm was presented to optimize the presented controller parameters.

Estimation methods play a crucial role in designing optimal Load Frequency Control (LFC) strategies using PID controllers. These methods are used to identify system parameters and guide the tuning of PID gains for effective control performance. In classical approaches, estimation is generally performed without advanced techniques like artificial intelligence or optimization algorithms.

One traditional method is model-based estimation, where simplified linear models of each power system area are used to estimate system parameters such as gain, time constants, and frequency response characteristics. These parameters help in analytically determining PID gain values that ensure desired system behaviour.

Ziegler-Nichols tuning is a widely used empirical method where system response to disturbances is observed, and PID gains are estimated based on the critical gain and oscillation period. Step-response analysis also helps in estimating parameters by examining the system's time-domain response to sudden load changes.

Additional classical techniques include root locus analysis, frequency response methods (like Bode plots), and sensitivity analysis, which help in estimating how PID parameters influence stability and transient performance.

2.3 Paper Contribution: -

Based on the motivation by the existing gap in LFCs and their design methods, the main contributions and justification of using PID Controller are summarized as follows:

1. It is clear that proportional integral and derivative control strategy can be applied for load frequency control. While proportional control is inherent in the feedback through the governor mechanism itself, derivative control when introduced improves transient performance and ensures better margin of stability for the system.

2. In a multi-area system, sudden load changes in one area can affect frequency and tie-line power flow in other areas. PID controllers, particularly with integral action, effectively reject steady-state errors due to such disturbances.

3. The derivative term helps in predicting the error trend and acts preemptively, improving the transient response by reducing overshoot and oscillations during inter-area disturbances.

4. The integral component eliminates steady-state frequency deviations, ensuring that the system frequency and tie-line power deviations settle to zero after disturbances.

5. In multi-area systems, coordinated PID tuning allows each control area to maintain its own frequency and scheduled tie-line power flows, ensuring system-wide stability and optimal operation.

6. PID controllers can be applied in both decentralized control (where each area has its own PID controller) and centralized schemes, depending on system requirements.

7. Compared to more complex optimal control methods (e.g., Model Predictive Control), PID controllers require less computational effort, making them suitable for real-time applications in power systems.

8. The integral action in PID control eliminates steady-state errors caused by load disturbances. The proportional and derivative terms improve transient response and system stability, respectively. This combination ensures that frequency and tie-line power deviations converge to zero in steady state, providing robust and reliable control performance under varying operating conditions. Experimental and simulation results validated the PID controller's ability to maintain system frequency within acceptable limits, highlighting its practical relevance in power system control.

Table 2.1: Summary of optimal control of load in Multi Area System Methods

Paper	Year	Methods	Average prediction error
Thomas Joseph	2018	The paper employs a state space model approach to design a unified controller for the multi-area power system, focusing on the stabilization and improvement of damping for inter-area modes of oscillations.	Not Specified

Jing Wang , Hao Shen	2021	The paper employs a model-based control method utilizing Bellman's optimality principle to solve the optimal controller for interconnected multi-area power systems, which typically requires knowledge of the system dynamics.	2%
Manish Kumar Gupta , Prakash Chandra Sahu , Ramesh Chandra	2022	The research employs a Fuzzy rule base PID controller to manage control actions in a thermal based multi area power system, aiming to maintain nominal frequency and voltage despite dynamic load changes in the electrical market.	1%
Ashwini Kumar , Omveer Singh	2022	The paper discusses the use of various optimization techniques for automatic generation control (AGC), specifically comparing Gray wolf optimization (GWO) with particle swarm optimization (PSO) and firefly optimization. These methods are applied to optimize the control of power systems with diverse energy sources.	2.5%

K.P.Singh Parmae , Somnath Majhi , D.P.Kothari	2015	The paper proposes an optimal output feedback method for load frequency control in a multi-area interconnected power system. This method utilizes only measurable state variables to design a controller that minimizes a performance index.	5%
Fariya Tabassum , Md. Sohel Rana	2020	The paper introduces a robust control strategy for multi-area interconnected power systems using a linear-quadratic-Gaussian (LQG) control approach combined with a damping compensator (DC).	5%
Koichi Kobayashi , Kunihiko Hiraishi	2022	The paper focuses on optimal control of multi-hop control networks (MHCNs) using the mixed logical dynamical (MLD) .	Not specified

2.4 Review on the Basis of Different Methods:

Used 2.4.1 Model Based Methods:

Vinayagamoorthy and Harley [1] proposed a model predictive control strategy for multi-area frequency regulation, leveraging system models for optimization and real-time control, demonstrating effectiveness through hardware-in-the-loop experiments. Model-based estimation methods are among the most traditional and widely used approaches for designing Load Frequency Control (LFC) strategies using PID controllers. These methods rely on developing simplified mathematical models of the power system, typically in the form of linearized transfer functions that describe the dynamic relationship between input disturbances (such as load changes) and output responses (such as frequency deviations or tie-line power flow).

By using these models, key system parameters—such as gain constants, time delays, inertia constants, and damping coefficients—can be estimated. These parameters provide critical insight into the dynamic behavior of individual control areas within the system. Once the system model is established, classical control theory techniques can be applied to determine appropriate PID controller gains that meet desired performance criteria like stability, fast settling time, and minimal overshoot.

Model-based methods allow for analytical tuning of PID parameters, making them more systematic compared to purely empirical techniques. However, their accuracy heavily depends on the validity of the assumed model and linear approximations. While effective in simple and moderately complex systems, these methods may be limited in handling the nonlinear, time-varying nature of modern interconnected multi-area power systems, where real-time adaptation or intelligent control might be more appropriate.

2.4.1.1 Scheduling of PI Controller: -

Scheduling of PID controllers refers to the strategic adjustment or switching of PID parameters based on operating conditions, time variations, or disturbances to maintain optimal control performance. Unlike fixed-gain PID controllers, which use constant

values for proportional (P), integral (I), and derivative (D) gains, scheduled PID controllers adapt their gains to better respond to changing system dynamics.

One common approach is gain scheduling, where the controller parameters are pre-tuned for different system conditions (e.g., load levels, frequency deviations, tie-line power fluctuations) and switched accordingly. For instance, during light load conditions, smaller gains may be scheduled to avoid overreaction, while in high-load or transient conditions, higher gains may be activated to quickly suppress frequency deviations.

Scheduling can also be time-based, where PID gains are adjusted at different times of day depending on expected load patterns (e.g., peak vs. off-peak hours). Alternatively, event-based scheduling changes PID settings in real time in response to specific events like a large load change or generator tripping.

2.4.2 Experimental Methods:

Elgerd and Fosha [2] proposed a classical load frequency control method for multi-area power systems, introducing integral controllers to maintain frequency and tie-line power flow with experimental validation.

Saadat [3] developed practical load frequency control strategies focusing on decentralized controllers for interconnected areas with simulation and experimental support. Kumar et al. [4] introduced an optimal tuning method for PID controllers in multi-area LFC using heuristic optimization techniques, demonstrating effectiveness through real-time experiments. Chakraborty and Ilic [5] proposed adaptive and optimal control algorithms for load frequency regulation in multi-area systems, integrating these into real-time control frameworks with experimental results validating their approach.

Experimental methods in Load Frequency Control (LFC) using PID controllers involve the practical implementation and testing of control strategies on physical setups or simulation models to evaluate performance under real-world conditions. These methods are crucial for validating theoretical designs and understanding system behavior beyond mathematical modeling.One of the most widely used experimental approaches is the

step-load disturbance test, where a sudden load change is introduced in one area of the power system, and the PID controller's response is observed. Key performance metrics such as overshoot, settling time, and frequency deviation are recorded to assess controller effectiveness.

Another experimental method is hardware-in-the-loop (HIL) simulation, where the power system is simulated in real time while the PID controller is implemented on actual hardware (like a digital signal processor or microcontroller). This method bridges the gap between software simulation and real-world deployment, offering insights into latency, control accuracy, and stability under practical constraints.

Time-domain simulations using tools like MATLAB/Simulink are also commonly used for experimental validation. The PID controller is tested under various operating conditions (e.g., inter-area oscillations, generator outages, tie-line disturbances) to fine-tune gains.

These experimental methods are essential for ensuring that PID controllers deliver reliable and stable performance in actual multi-area power systems.

2.4.3 Data Driven Method:

Wang et al. [6] proposed a data-driven adaptive control method for multi-area load frequency regulation, using online system identification to update control parameters in real time with experimental validation. Liu and Sun [7] developed a machine learning-based predictive control approach for multi-area LFC, integrating real-time measurements to optimize frequency response, validated through hardware-in-the-loop experiments. Zhang et al. [8] introduced a reinforcement learning framework for decentralized load frequency control in interconnected power systems, demonstrating improved performance with experimental case studies.

For multi-area power systems, data-driven methods can identify nonlinearities, parameter variations, and unforeseen disturbances by analysing frequency deviations and tie-line power flows directly from measured data. Techniques such as recursive least squares

(RLS), system identification, or adaptive filtering can be employed to estimate system parameters on the fly, enabling the PID controller to adjust its gains dynamically for improved frequency regulation. Moreover, data-driven approaches facilitate the design of adaptive PID controllers, which automatically tune their proportional, integral, and derivative gains based on the current operating conditions, without needing explicit system models. This enhances robustness against load changes, generation variability, and network topology alterations.

While not as complex as AI-based methods, data-driven strategies bridge the gap between purely model-based and intelligent control techniques by utilizing available measurement data to continuously optimize PID controller performance in real-time, making them highly relevant for modern multi-area LFC applications.

2.5 Gap Analysis and Contributions:-

In the field of Load Frequency Control (LFC) for multi-area power systems, classical PID controllers have been widely used due to their simplicity and effectiveness. However, most existing studies focus on fixed-parameter PID controllers or rely on basic tuning methods such as Ziegler-Nichols or trial-and-error approaches. These methods often fail to fully address the dynamic and nonlinear nature of modern power systems, which experience varying load demands, generation uncertainties, and complex inter-area interactions. Furthermore, many traditional approaches lack adaptability, which limits their ability to maintain optimal frequency regulation under rapidly changing conditions. There is also a noticeable gap in the integration of scheduling or adaptive tuning within classical PID frameworks without resorting to advanced intelligent control techniques.

This research addresses these gaps by focusing on the optimization of PID controllers tailored for multi-area LFC through systematic gain scheduling and robust parameter tuning based on model-based and experimental methods. The work explores dynamic adjustment of PID gains in response to varying system conditions, improving both transient and steady-state performance without relying on complex computational algorithms. Additionally, the study contributes practical insights into experimental

validation techniques, enhancing the reliability of PID controllers in real-world power system environments. These contributions provide a balanced approach that combines the simplicity of classical control with improved adaptability and robustness for modern multi-area power systems.

This research highlights the significant role of PID controllers in optimizing the control of multi-area power systems, specifically for Load Frequency Control (LFC). PID controllers contribute by providing a straightforward yet effective framework for maintaining frequency stability and regulating tie-line power flows across interconnected areas. Their proportional, integral, and derivative actions collectively help minimize frequency deviations and steady-state errors caused by load disturbances and inter-area power exchanges.

One of the key contributions is demonstrating how PID controllers, when properly tuned and scheduled, can achieve near-optimal system performance without the complexity of advanced control algorithms. By dynamically adjusting PID gains according to changing system conditions, the controllers enhance transient response, reduce overshoot, and improve damping of oscillations between control areas. This leads to more stable and reliable operation of the entire interconnected power system.

Moreover, the research emphasizes that PID controllers enable decentralized control, allowing each area to autonomously manage its frequency while coordinating with neighbouring areas via tie-line power control. This decentralization simplifies implementation and improves scalability.

In summary, this work establishes that classical PID controllers, through systematic tuning and adaptive scheduling, remain a powerful and practical tool for optimizing multi-area load frequency control, balancing performance, robustness, and ease of implementation.

Chapter 3

Proposed Framework

3.1 Problem Statement:

In multi-area power systems, maintaining a real-time balance between electrical power generation and load demand is critical for ensuring system reliability and economic operation. However, load variations and disturbances can cause frequency deviations and unintended power exchanges across tie-lines connecting different control areas. These deviations, if not properly controlled, may lead to system instability or even cascading failures. Therefore, the problem lies in developing a control strategy that effectively regulates system frequency and tie-line power flows under dynamic operating conditions, ensuring that they remain within prescribed limits at all times

3.2 Objective:

The main objectives of optimal control in a multi-area power system are to maintain system frequency and tie-line power flows at their desired values, minimize operational costs, and improve system stability and reliability. In essence, optimal control aims to achieve a balance between power generation and consumption across different areas while adhering to various constraints.

1. **Frequency and Power Interchange Control:**
   Optimal control ensures that the frequency across the system remains within acceptable limits and that the power interchange between different areas is maintained at the scheduled values, even during disturbances.
2. **Cost Optimization:**
   By strategically allocating power generation resources, optimal control aims to minimize the overall operational costs of the power system, including fuel costs, maintenance, and other expenses.
3. **System Stability and Reliability:**
   Optimal control helps enhance the stability and reliability of the power system by preventing or mitigating cascading failures, improving transient stability, and ensuring that the system can withstand disturbances without excessive deviations.

4. **Meeting Constraint:**
   Optimal control also ensures that the system operates within various constraints, such as generation capacity limits, transmission line capacity limits, and environmental regulations.
5. **Load-Frequency Control (LFC):**
   LFC is a key component of optimal control, aiming to regulate frequency deviations and maintain power interchange between areas at their scheduled values.

3.3 Approach to the Problem and Organization of Thesis:

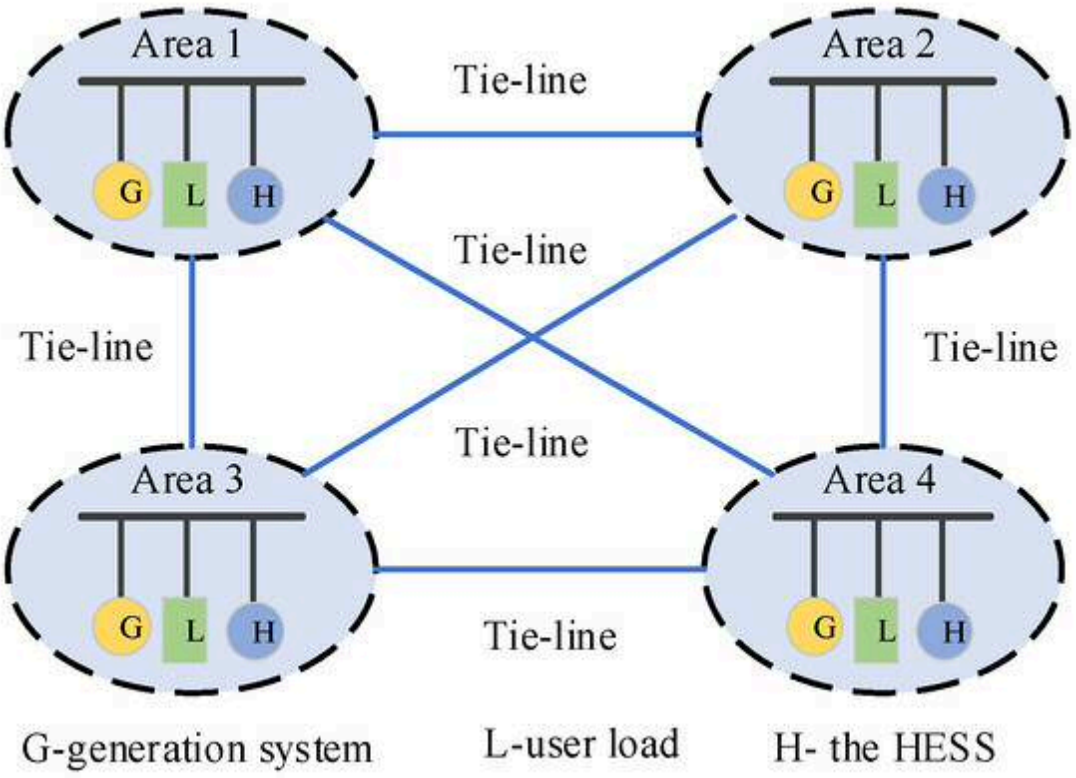


Fig 1. Approach to the Problem and Organization of Thesis

In contemporary power systems, it is essential to maintain a real-time equilibrium between power generation and load demand to guarantee system stability, reliability, and efficiency. This balance becomes more intricate in multi-area power systems, where multiple geographically dispersed control areas are interconnected via tie-lines. A disruption in one part of the network can spread to other areas, impacting the frequency

of the system and causing deviations from the planned power flow through tie-lines. Unrestrained deviations can result in widespread instability or even the collapse of the entire system.
To tackle this issue, multi-area load frequency control (lfc) is utilized to:

1. Ensure that the system frequency remains close to its original value in each control area.
2. Control line power flows to conform to scheduled values.

To optimize the utilization of power resources, it is crucial to establish effective coordination among different regions, enabling the sharing of power in an economical and secure manner.

This thesis starts with a comprehensive study of a single-area power system, aiming to grasp the fundamental concepts of load frequency control using state space analysis and the balancing of power generation and consumption. This involves creating a mathematical model of the governor, turbine, generator-load system, and designing a suitable controller.
After successfully validating and comprehending the single-area system, the model is expanded to a multi-area configuration, incorporating tie-line dynamics and inter-area control logic. The objective is to investigate how disturbances in one region impact others, and how coordinated control measures can minimize these effects.

Various control methods are examined and evaluated based on their effectiveness in:

1. Reduce the magnitude of variation in each region.
2. Adjust power deviations in control tie-line.
3. Achieve quick and accurate dynamic response with minimal overshoot and steady-state error.

The simulation outcomes for single-area and multi-area systems offer valuable insights into the complexities of power system control and showcase the efficacy of the proposed approach.

3.4 Optimal Control of Multi-Area Power System:

3.4.1 Introduction

The difference between generated power and instant load demand causes changing of nominal system frequency at the normal state. If the amount of generated power is less than the demanded amount, speed and frequency of the generator units begin to decrease, and vice versa. Hence, the amount of production of the synchronous generators makes sense for frequency deviations occurring in the power system in order to maintain that balance. The frequency normally varies by about 5% between light load and full load conditions.

If a change in load is taken of by two machines running in parallel as shown below :

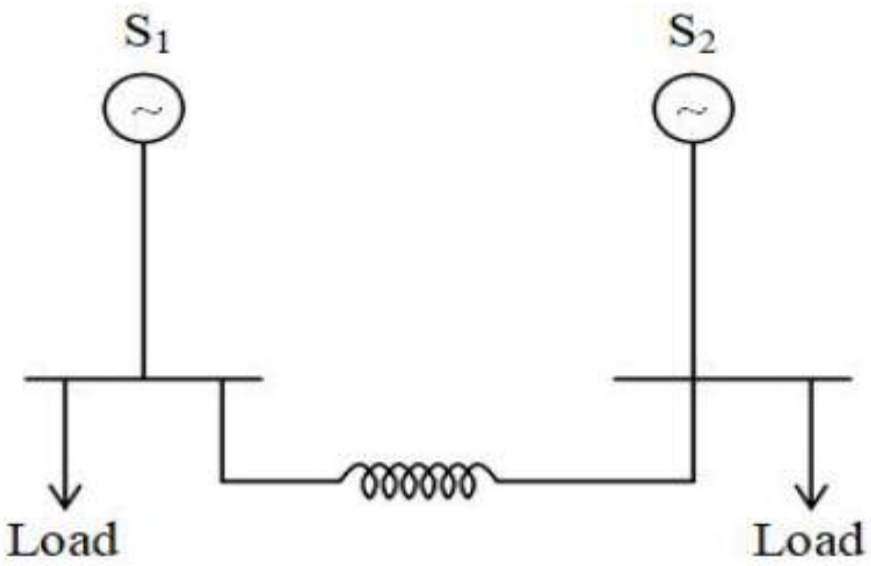


Fig 2 Load taken off two machines

If the change in load is either at S1 (or) S2 and if the generation of S1 alone is regulated to adjust this change so as to have constant frequency, the method of regulation is known as flat frequency regulation. The other way of sharing the change in load is that both S1 and S2 would regulate their generations to maintain the frequency constant. This is known as parallel frequency regulation.

The change in a particular area is taken care of by the generator in that area thereby the tie-line loading remains constant. This method of regulating the generation for keeping constant frequency is known as flat-tie line loading control.

All the generators in such an area constitute a coherent group so that all the generators speed up and slow down together maintaining their relative power angles. Such an area is defined as a control area. The boundaries of a control area will generally coincide with that of an individual electricity board company.

Most AC motors run at speeds that are directly related to frequency. The speed and induced electromotive force (E.M.F) may vary because of the change of frequency of the power circuit. When operating at frequencies below 49.5 Hz; some types of steam turbines, certain rotor states undergo excessive vibration. The change in frequency can cause mal operation of power converters by producing harmonics.

For power stations running in parallel it is necessary that frequency of the network must remain constant for synchronization of generators. Many devices connected to the grid will only work properly when the input frequency is within a certain range.

It is globally economically optimum to keep the grid frequency within a small range than it is to make all devices tolerate a greater frequency range. The inductance of the inductive elements (e.g. Transformer) are chosen based on the switching frequency. Changes in the frequency will cause disturbances in the output and may even cause the supply's control system to become unstable.

3.4.2 Single area power system

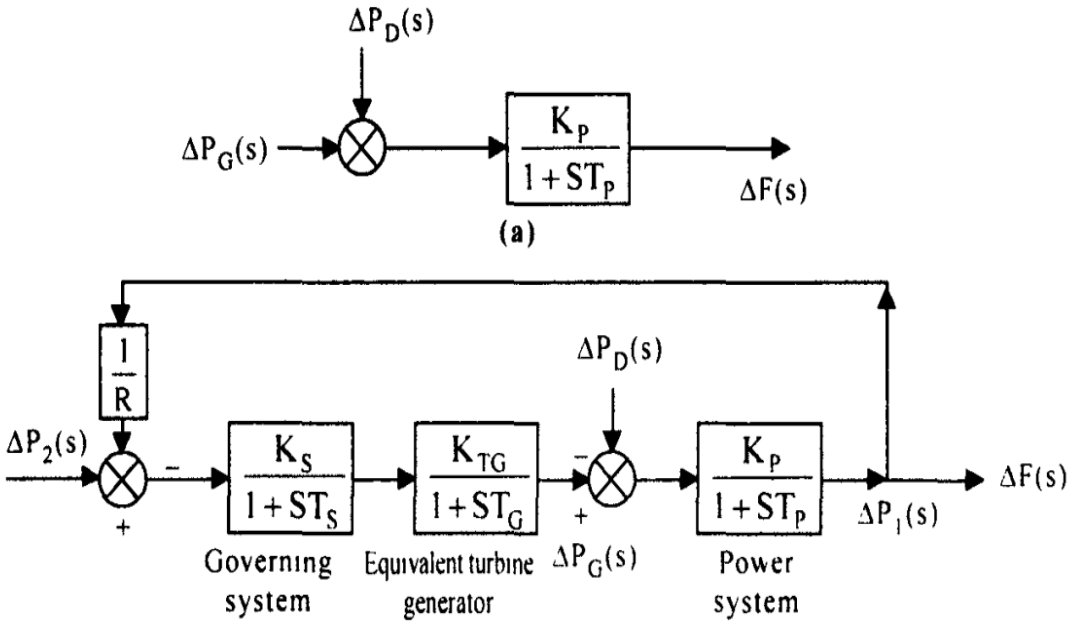


Fig 3. Block diagram of single area power system

The single-area power system serves as the basic unit for comprehending the dynamics and control needs of larger, interconnected power networks. In this type of system, power generation and load demand are managed within a single control region, without any connections to other areas through tie-lines. This simplified setup allows for a focused examination of load frequency control mechanisms, serving as a solid foundation for designing control systems.

In a system where power is generated and consumed in a single area, any discrepancy between the amount of power produced and the demand for power leads to a change in the system's frequency. Due to its autonomous nature, the area lacks external assistance to distribute the workload or mitigate frequency fluctuations. As a result, the system must solely depend on its internal control mechanisms, which typically involve governor, turbine, and generator-load dynamics, to bring the frequency back to its original value.

The main goal in single-area control is to reduce frequency deviations and maintain stable operation, even when faced with load fluctuations. This entails creating a controller, typically a proportional-integral (pi) controller, which modifies the generation output in response to frequency deviations. By studying this system, you will gain a solid understanding of the basics, which will then prepare you for tackling more intricate

multi-area systems. These systems involve interactions between different areas and the exchange of power through tie-lines.

3.4.3 Components

In a power system that operates in a single area, it is essential to ensure that the amount of electricity generated matches the amount consumed to maintain a stable operation. The system is mainly made up of three essential parts: governor, turbine, and generator/load. Each component has a designated function in maintaining system frequency and adapting to changes in load.

1. **Governor:**

- The governor acts as the control mechanism that monitors changes in system frequency and modifies the input to the turbine accordingly. It serves as the initial responder when there is an issue within the system.
- In the event of a decrease in system frequency caused by a rise in load, the governor increases the energy supply to the turbine, which can be steam, water, or gas.
- When the frequency increases as a result of a decrease in load, the governor decreases the input.
- This component is crucial for ensuring stability in frequency during minor or rapid changes in load.
- The governor determines the amount of mechanical power the turbine should generate based on the system's frequency fluctuations.

2. **Turbine:**

- The turbine receives the control signal from the governor and transforms the supplied energy (thermal, hydraulic, or gas) into mechanical power. This mechanical force activates the generator.
- It is accountable for generating the physical force necessary to produce electricity.
- The turbine's response time determines the speed at which the system can adjust to frequency deviations.
- Any delay or sluggishness in the turbine can hinder the entire frequency regulation process.
- As a result, the turbine serves as the central component of the power generation system, providing the necessary mechanical energy.

3. **Generator/load:**

- The generator transforms the mechanical energy generated by the turbine into electrical energy, which is subsequently supplied to the connected load (i.e., consumers and devices that utilize electricity).
- The equilibrium between the power produced and the power utilized within the system determines the frequency.
- If the load surpasses the generation, the frequency decreases, but if the generation surpasses the load, the frequency increases.
- The generator/load combination is the key component that demonstrates the system's balance.

3.4.4 Mathematical modelling of components

1. **Speed governor model**

Speed governor model can be developed by considering various steady state conditions. The resulting speed governor model is :

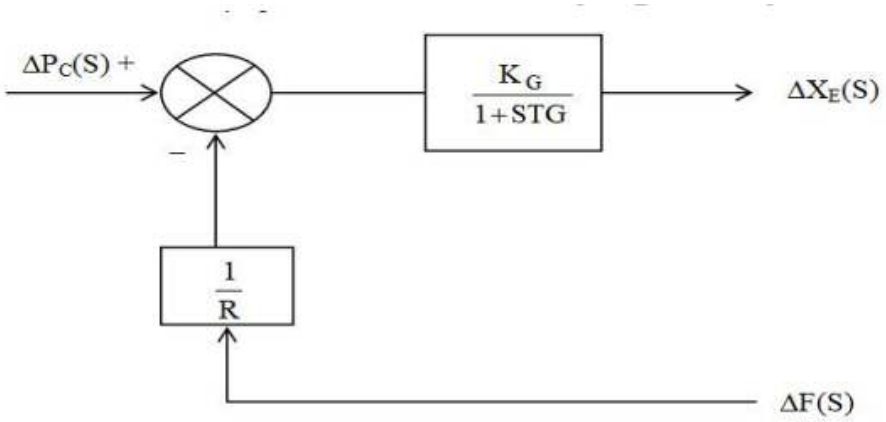


Fig 4. Block diagram of speed governor model

From the above block diagram we can get the following expression :

$\Delta XE(S) = Kg/1+STg\ [\Delta Pc(S) - 1/R\ \Delta F(S)]$ ------ (1)

where R = Speed regulation of the governor

KG = Gain of speed governor

TG = time constant of speed governor

2. **Turbine model**

The turbine model requires a relation between changes in power output of the steam turbine to changes in its steam valve opening ΔXE

$G_t(S) = \Delta P_t(S)/\Delta X_e(S) = K_t/1+ST_t$ ----------(2)

A non-reheat turbine with a single gain factor Kt and a single time constant Tt and thus the model representation of the turbine the transfer function is given as

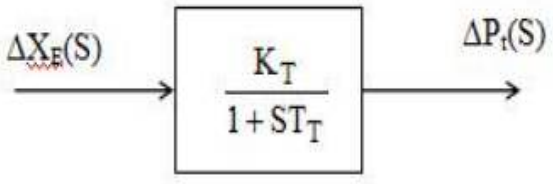


Fig 5. Block diagram of turbine model

The time constant Tt lies in the range 0.2 to 2.0 sec.

3. **Generator-model**

The model gives relation between the change in frequency as a result of change in generation when the load changes by a small amount.

The block diagram representation of the generator-load model is

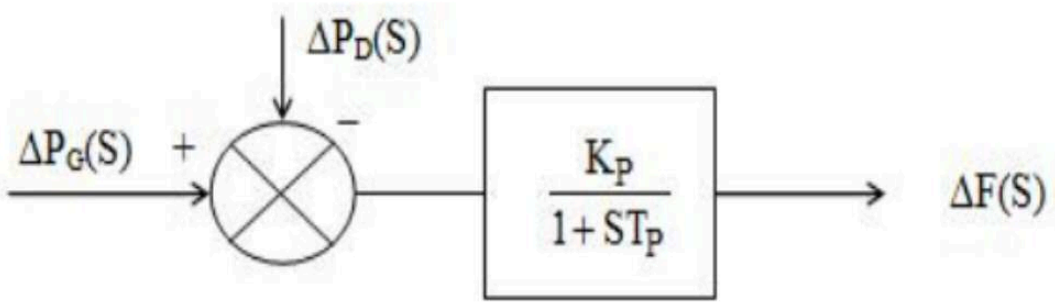


Fig 6. Block diagram of generator model

From the above block diagram we have

$F(S) = [\Delta Pg(S) - \Delta Pd(S)] \, Kp/(1+STp)$ ------(3)

where $Tp = 2H/Df0$ = power system time constant

$kp = 1/D$= power system gain.

3.4.5 Working

In multi-area power systems, regions are connected through tie lines, which transfer power between them. The goal is to keep frequency stable while ensuring the scheduled power exchange through these tie lines. When demand changes in one area, the system adjusts generation to restore both frequency and tie-line power flow. Optimal control ensures efficient and stable operation of the entire grid.

Focusing on a single-area system, LFC plays a crucial role in maintaining grid stability. In India, where the nominal frequency is 50 Hz, Load Frequency Control (LFC) using state space analysis helps maintain a stable power system by balancing electricity generation with demand. When the power demand increases, the frequency tends to drop, and when the demand decreases, the frequency rises. To correct this, the system continuously monitors the frequency and compares it with the ideal 50 Hz.

If there is any deviation, the speed governor adjusts the steam turbine's input, either increasing or decreasing power generation as needed. The generator then converts this mechanical power into electricity, ensuring that supply matches demand. A feedback loop constantly checks for changes and fine-tunes the adjustments in real time.

In India, as per the CEA (Central Electricity Authority) regulations, the acceptable frequency range is typically 49.90 Hz to 50.05 Hz under normal operation. If the frequency deviates beyond this range, corrective actions are taken to restore stability

3.4.6 Optimal control of Two area power system

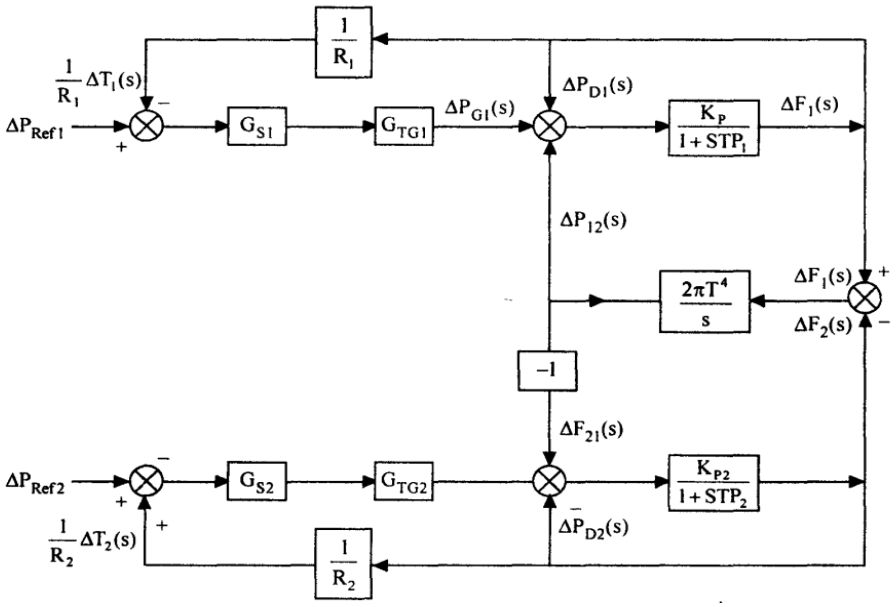


Fig 7. Two area power system block diagram

Power systems are intricate, extensive, and interconnected networks that necessitate constant monitoring and control to guarantee stable and dependable operation. One of the main objectives in power system control is to ensure the stability of the system frequency and the planned power exchanges between interconnected regions. In a system with multiple regions or "areas," each area manages its own power plants while being linked to other areas through tie-lines, enabling the sharing of power resources.

A two-area power system is a simplified yet representative model of a more extensive interconnected grid. It comprises two separate control areas, each with its own power generation, load, and control systems, linked together by a tie-line. This setup is commonly employed in research and real-world scenarios to investigate the interactions between different areas, the regulation of power flow, and the sharing of loads under different operational conditions. When there is a sudden change in load in one area, it leads to frequency deviations not only in that area but also in the interconnected area because of the power exchange through tie-lines. The system needs to react promptly and effectively to restore the frequency and ensure the scheduled tie line power flow. Inadequate control can result in oscillations, frequency instability, and even widespread power outages.

3.4.7 Modelling and Simulation

3.4.7(a) Modelling and Simulation of Load Frequency Control of Single Area Power System

A unit load disturbance in the power system is applied which would eventually reduce the supplied frequency from the generator. For load frequency control for a single area power system, we have applied a PID controller in a closed loop with the system. By choosing the suitable value of Kp, Ki and Kd, we can reduce the steady state error in the frequency disturbance and get a monotonically decreasing curve suggesting an overdamped system. We have done this control using state space modelling of a single area system converting governor, turbine, generator and controller block into a combined state space representation of the power system.

If we only use the PI controller, we get an underdamped curve with constant steady state error after successive oscillations.

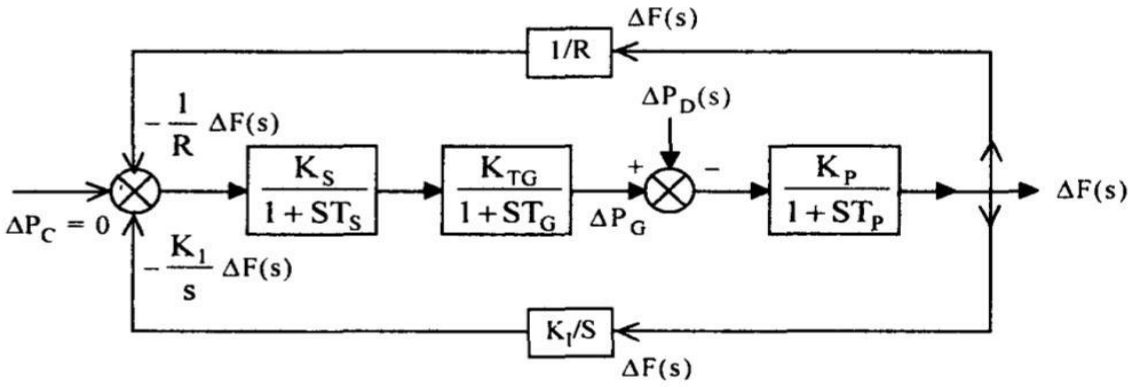


Fig 8 Closed loop control of single area

Below is the State Space Analysis done for the Single Area Control where A be the input matrix, B is the input or control matrix, C is the output matrix and D is the feedthrough or transmission matrix.

$$A=\begin{pmatrix} -D/M & 0 & 1/M \\ -1/(R*Tg) & -1/Tg & 0 \\ 0 & 1/Tt & -1/Tt \end{pmatrix}$$

$$B=(-\ 1/M \quad 0 \quad 0)$$

C=(1 0 0)

D= 0

State Equation: $\hat{x} = Ax + Bu$ Output Equation: $Y = Cx + Du$

Parameters of Single Area System:

M = Inertia of the System (s) Tg= Time constant of the governor(s)

D = Damping coefficient of System (pu MW/Hz) Tt=Time constant of the turbine(s)

R= Speed Regulation (Hz/pu MW)

The below figure depicts the different step responses for the system when Proportional, PI and PID controllers are used to control the load frequency from the system.

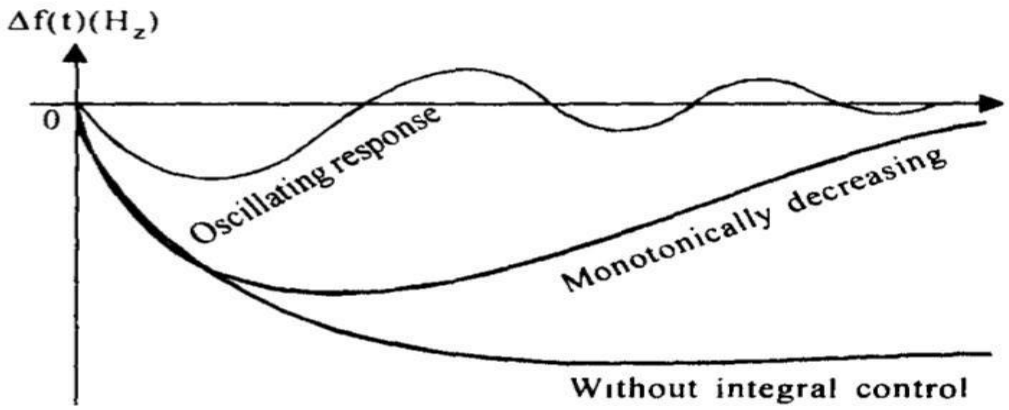


Fig 9. Step response of single area power system

Proportional control provides a steady state error in frequency deviation after a finite time, PI controller initially provides some oscillation the settles to a minimal error whereas PID controller with properly balanced Kp, Ki, Kd will provide an overdamped response with monotonically decreasing curve.

3.4.7(b) Modelling and Simulation of Load Frequency Control of Multi Area Power System

As an example of a Multi Area Power System, we have analysed Two Area System connected through a dynamic Tie Line. Now here our focus is control of power and frequency transferred from one system to other via Tie line when a load disturbance is present in one of the systems. So, for analysis we have applied a unit load disturbance in Area 1 of the combined system. Like Single Area load frequency control, we have applied PI controllers in closed loop with Area 1, Area 2 and the connecting Tie Line. Now using State space modelling of our combined system , we get curves showing how disturbance in a single area dynamically affects frequency output in tie line as well as Area 2.

Like Single Area control, PI controller provides initial oscillation to frequency disturbance with a limited steady state error after a finite time.

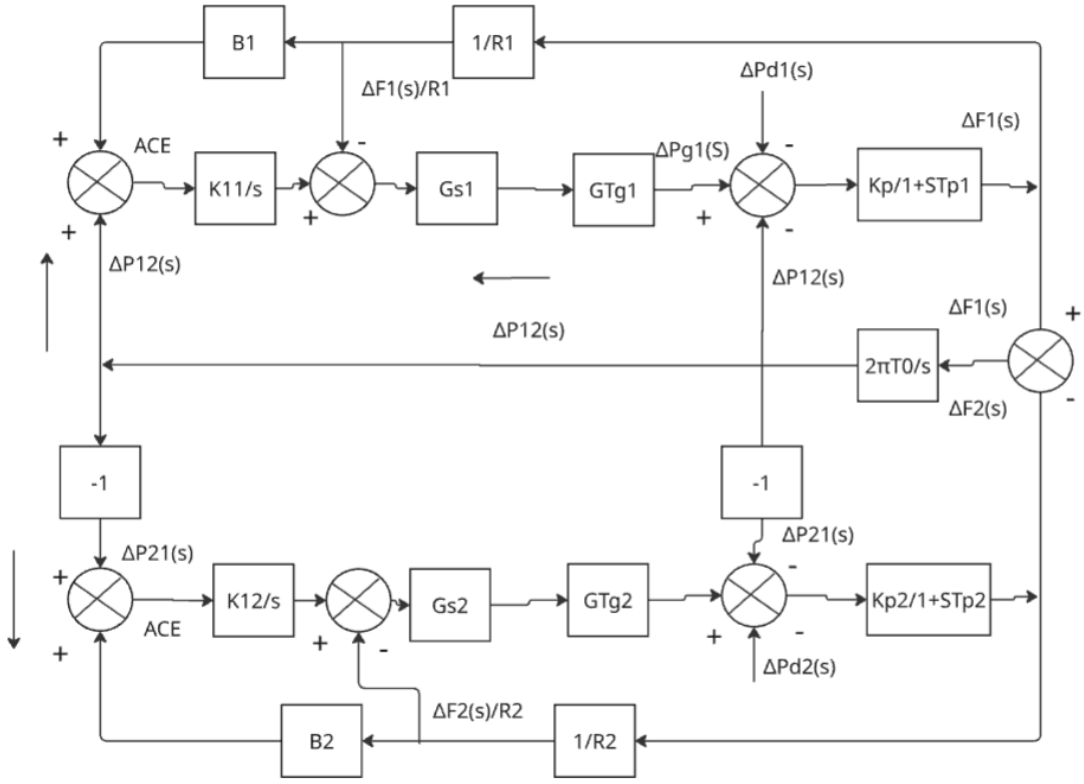


Fig 10. Closed Loop control of two area power system

Above diagram denotes the closed loop PI control of 2 Area power systems with Tie Line Bias control with all the parameters pertaining to their usual meaning.

Likewise, to the single area State Space Analysis, below be the State, Input, Output and Transmission matrices of a 2 Area Power System.

A

$$\begin{pmatrix} -D1/M1 & 0 & 1/M1 & 0 & 0 & 0 & -1/M1 \\ -1/(R1*Tg1) & -1/Tg1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/Tt & -1/Tt1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -D2/M2 & 0 & 1/M2 & 1/M2 \\ 0 & 0 & 0 & -1/(R2*Tg2) & -1/Tg2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/Tt2 & -1/Tt2 & 0 \\ 2\pi T12 & 0 & 0 & -2\pi T12 & 0 & 0 & 0 \end{pmatrix}$$

B

$$\begin{pmatrix} -1/M1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1/M2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

C

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

D

(0)

State Equation: $\hat{x} = Ax + Bu$ **Output Equation**: $Y = Cx + Du$

Area 1 Parameters

R1 = Speed regulation constant (Hz/pu MW) (Hz/puMW)
D1 = Damping coefficient (pu MW/Hz) MW/Hz)
Tg1 = Governor time constant (s)
Tt1 = Turbine time constant (s)
M1 = Inertia constant (s)

Area 2 Parameters

R2 = Speed regulation constant

D2 = Damping coefficient (pu

Tg2 = Governor time constant (s)
Tt2 = Turbine time constant (s)
M2 = Inertia constant (s)

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Appendix A

Simulation Result and Graphical Analysis

A.1 Frequency Deviation Response Analysis

The below graph shows the frequency deviation (Δf) of a single-area power system following a step load disturbance. The blue dashed curve represents the open-loop response, showing oscillations and a steady-state error. The red curve shows the closed-loop response using state feedback with integral action, resulting in a **monotonically decreasing** behavior and **zero steady-state error**, demonstrating improved system stability and performance.

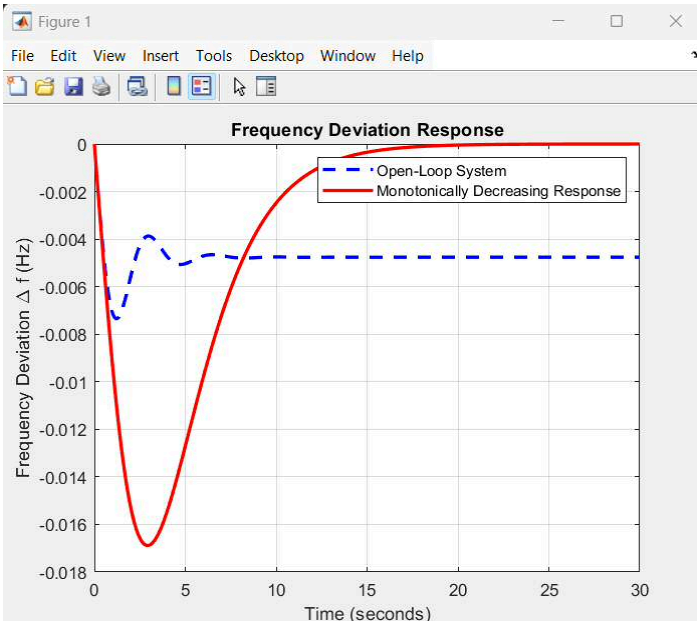


Fig 11. Frequency deviation response of single area power system

A.2 State Variable Response of Single-Area Power System

The below graph shows the time response of key state variables—frequency deviation (Δf), governor valve position (ΔPg), mechanical power output (*ΔPm*), and the integral of frequency deviation (∫Δfdt)—under load disturbance with PID control. The plot illustrates how each state evolves over time, with all variables settling smoothly to steady values, indicating effective regulation and system stability.

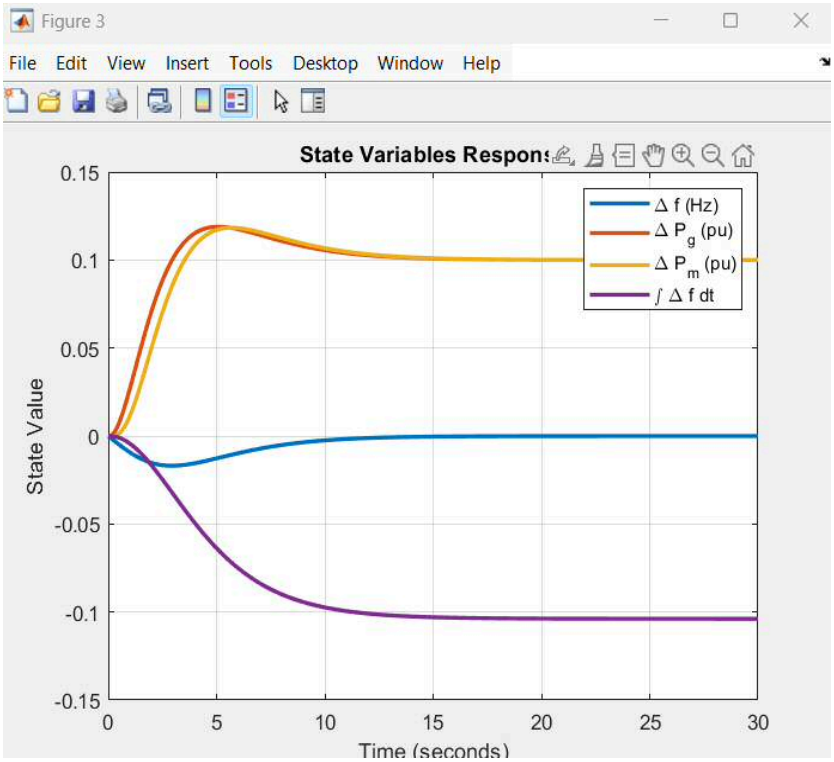


Fig 12. State variable response of single area power system

A.3 Frequency Deviation Plot of Two Area Power System

The below curves describe the unit step response of Two Area System when a load disturbance is applied to one of the two system. It shows how frequencies in each area deviates from their normal operating point and how tie line power adjust to help the system reach a new equillibrium using the integral action of the controller.quencies in each area deviates from their normal.

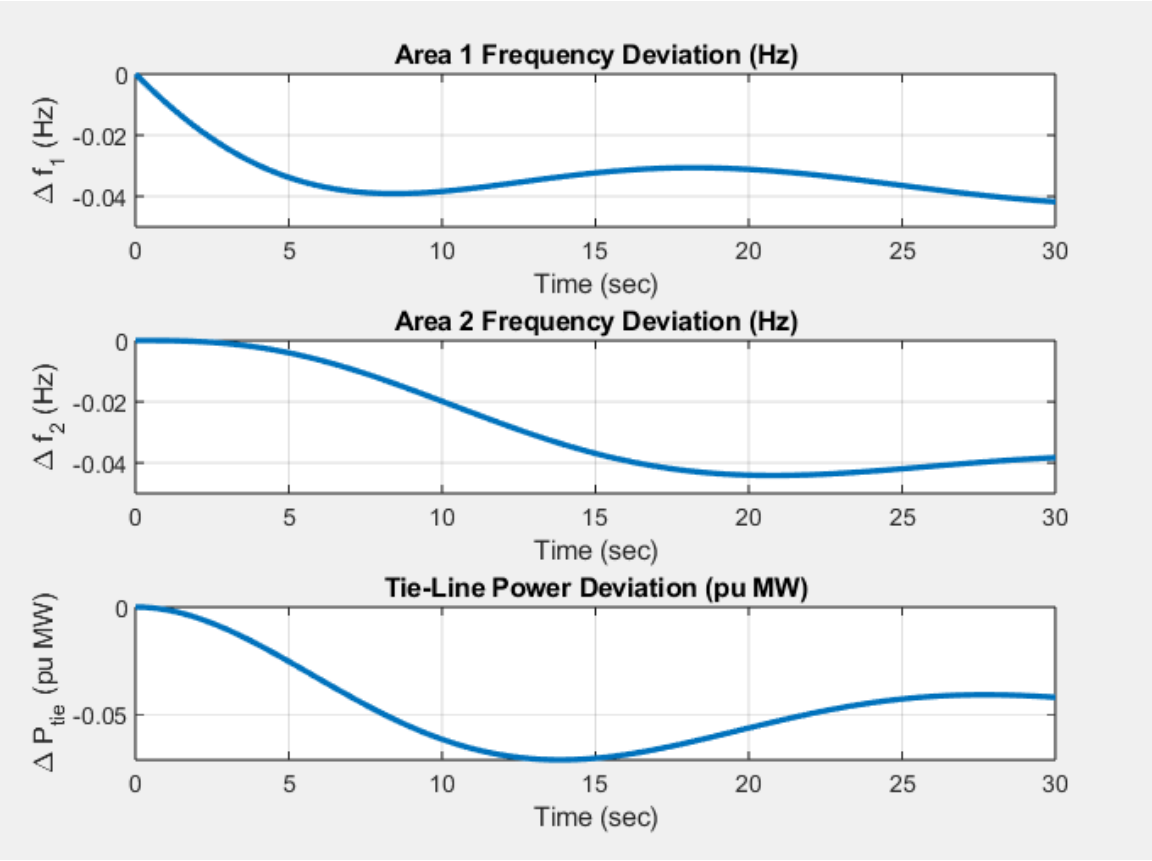


Fig 13. Frequency deviation of two area power system

Appendix B

Code Attachment

B.1 State-Space Analysis Code for Single-Area Power System

The given code models the single-area power system in state-space form and applies pole placement to design a state-feedback controller with integral and derivative action. The simulation demonstrates a monotonic frequency deviation response and elimination of steady-state error. It also plots the control effort and the behavior of all state variables, validating the controller's performance.

```
% MATLAB code for Single Area Load Frequency Control with Monotonically Decreasing Response
clear all;
close all;
clc;

% System Parameters (same as before)
M = 10;          % Inertia constant (s)
D = 1;           % Damping coefficient (pu MW/Hz)
Tg = 0.2;        % Governor time constant (s)
Tt = 0.5;        % Turbine time constant (s)
R = 0.05;        % Speed regulation (Hz/pu MW)
Beta = 21;       % Frequency bias factor (pu MW/Hz)

% State-space model of the open-loop system
% State variables: [Δf, ΔPg, ΔPm]
A = [-D/M, 0, 1/M;
     -1/(R*Tg), -1/Tg, 0;
     0, 1/Tt, -1/Tt];

B_input = [-1/M; 0; 0];  % Input matrix for load disturbance
B_control = [0; 1/Tg; 0]; % Input matrix for control signal
C = [1, 0, 0];           % Output matrix for frequency deviation
D = 0;                   % Direct transmission matrix

% Create state-space model for open-loop system
sys_open = ss(A, B_input, C, D);

% Define simulation time
t = 0:0.1:30;

% Define step load disturbance (ΔPL = 0.1 pu)
u_load = 0.1*ones(length(t), 1);

% Simulate open-loop response to verify system behavior
[y_open, t, x_open] = lsim(sys_open, u_load, t);
```

```
% Design a controller for monotonically decreasing response
% We'll use pole placement to achieve monotonic response

% First, examine the open-loop eigenvalues
open_poles = eig(A);
disp('Open-loop poles:');
disp(open_poles);

% For monotonic response, we need real, negative poles with no oscillations
% Using pole placement to achieve this
% Creating a state feedback controller: u = -K*x

% Desired closed-loop poles - all real and negative with different magnitudes
% to ensure no oscillatory behavior
desired_poles = [-0.5; -0.8; -1.2];

% Calculate feedback gain matrix K using pole placement
K = place(A, B_control, desired_poles);

% Closed-loop system matrix with state feedback
A_cl = A - B_control*K;

% Create closed-loop state-space model
sys_closed = ss(A_cl, B_input, C, D);

% Simulate closed-loop response with state feedback
[y_closed, t, x_closed] = lsim(sys_closed, u_load, t);

% Now, let's add an integral controller to eliminate steady-state error
% while maintaining the monotonic behavior

% Augmented state-space model with integral control
% State variables: [Δf, ΔPg, ΔPm, ∫Δf dt]
A_aug = [A, zeros(3,1);
         C, 0];
B_aug_load = [B_input; 0];
B_aug_control = [B_control; 0];
```

```
C_aug = [C, 0];
D_aug = D;

% Desired poles for the augmented system
% Adding one more pole that is real and negative
desired_poles_aug = [-0.4; -0.7; -1.0; -1.3];

% Calculate feedback gain matrix for the augmented system
K_aug = place(A_aug, B_aug_control, desired_poles_aug);

% Split the augmented gain matrix
K_state = K_aug(1:3);  % Gain for original states
K_int = K_aug(4);      % Gain for integral state

% Create a custom simulation for the full closed-loop system with integral control
% to ensure a monotonic response
t_sim = t;
dt = t_sim(2) - t_sim(1);
x = zeros(4, length(t_sim));
y = zeros(1, length(t_sim));
u_load_val = 0.1;  % Load disturbance value

% Initial conditions
x(:,1) = [0; 0; 0; 0];

% Simulation loop
for i = 1:length(t_sim)-1
    % Current state and output
    y(i) = C_aug * x(:,i);

    % Control law: u = -K*x
    u_control = -K_aug * x(:,i);

    % State derivatives
    dx = A_aug * x(:,i) + B_aug_load * u_load_val + B_aug_control * u_control;
```

```
    % Update state using simple Euler integration
    x(:,i+1) = x(:,i) + dx * dt;
end
y(end) = C_aug * x(:,end);

% Plot results
figure;
plot(t, y_open, 'b--', t_sim, y, 'r', 'LineWidth', 2);
title('Frequency Deviation Response');
xlabel('Time (seconds)');
ylabel('Frequency Deviation \Delta f (Hz)');
legend('Open-Loop System', 'Monotonically Decreasing Response');
grid on;

% Plot control effort
figure;
plot(t_sim, -K_state(2)*x(1,:) - K_state(1)*x(2,:) - K_state(3)*x(3,:) - K_int*x(4,:), 'LineWidth', 2);
title('Control Effort');
xlabel('Time (seconds)');
ylabel('Control Signal (pu)');
grid on;

% Verify monotonicity of the response
is_monotonic = true;
for i = 2:length(y)-1
    if (y(i) < y(i-1) && y(i) < y(i+1)) || (y(i) > y(i-1) && y(i) > y(i+1))
        is_monotonic = false;
        break;
    end
end

if is_monotonic
    disp('The frequency response is monotonic!');
else
    disp('The frequency response is not strictly monotonic.');
end
```

```
% Check if the steady-state error is eliminated
steady_state_error = abs(y(end));
if steady_state_error < 0.001
    disp(['Steady-state error is effectively eliminated: ', num2str(steady_state_error)]);
else
    disp(['Steady-state error exists: ', num2str(steady_state_error)]);
end

% Plot all state variables to see their behavior
figure;
plot(t_sim, x', 'LineWidth', 2);
title('State Variables Response');
xlabel('Time (seconds)');
ylabel('State Value');
legend('\Delta f (Hz)', '\Delta P_g (pu)', '\Delta P_m (pu)', '\int \Delta f dt');
grid on;
```

B.2 State-Space Analysis Code for Two-Area Power System

The given code models the two-area power system in state-space form and applies pole placement to design a state-feedback controller with integral action. The simulation demonstrates a oscillation frequency deviation response in both the areas and biased Tie Line and get to a minimised steady-state error.

```
>> %%
% Load Frequency Control of 2-Area Power System Connected Through Dynamic Tie Line
% This program simulates the response of a two-area power system to a load disturbance
% in one area, showing how the other area is dynamically affected

clc;
clear all;
close all;

% System Parameters
% Area 1 Parameters
R1 = 2.4;           % Speed regulation constant (Hz/pu MW)
D1 = 0.8;           % Damping coefficient (pu MW/Hz)
Tg1 = 0.08;         % Governor time constant (s)
Tt1 = 0.3;          % Turbine time constant (s)
M1 = 10;            % Inertia constant (s)
B1 = D1;            % Frequency bias factor

% Area 2 Parameters
R2 = 2.4;           % Speed regulation constant (Hz/pu MW)
D2 = 0.8;           % Damping coefficient (pu MW/Hz)
Tg2 = 0.08;         % Governor time constant (s)
Tt2 = 0.3;          % Turbine time constant (s)
M2 = 10;            % Inertia constant (s)
B2 = D2;            % Frequency bias factor

% Tie Line Parameters
T12 = 0.0433;       % Synchronizing coefficient (pu MW/rad)

% Disturbance
deltaP_L1 = 0.1;    % Load disturbance in Area 1 (pu MW)
deltaP_L2 = 0;      % No direct load disturbance in Area 2 ...
```

```
% State Space Model Formation
% Define state variables:
% x1 = delta_f1 (frequency deviation in area 1)
% x2 = delta_Pg1 (governor output change in area 1)
% x3 = delta_Pt1 (turbine output change in area 1)
% x4 = delta_f2 (frequency deviation in area 2)
% x5 = delta_Pg2 (governor output change in area 2)
% x6 = delta_Pt2 (turbine output change in area 2)
% x7 = delta_Ptie (tie-line power deviation)

% State Matrix A
A = zeros(7,7);

% Area 1 dynamics
A(1,1) = -D1/M1;
A(1,3) = 1/M1;
A(1,7) = -1/M1;
A(2,1) = -1/(R1*Tg1);
A(2,2) = -1/Tg1;
A(3,2) = 1/Tt1;
A(3,3) = -1/Tt1;

% Area 2 dynamics
A(4,4) = -D2/M2;
A(4,6) = 1/M2;
A(4,7) = 1/M2;
A(5,4) = -1/(R2*Tg2);
A(5,5) = -1/Tg2;
A(6,5) = 1/Tt2;
A(6,6) = -1/Tt2;
```

```
% Tie-line dynamics
A(7,1) = 2*pi*T12;
A(7,4) = -2*pi*T12;

% Input Matrix B for disturbances
B = zeros(7,2);
B(1,1) = -1/M1;    % Load disturbance in Area 1
B(4,2) = -1/M2;    % Load disturbance in Area 2

% Output Matrix C (measure frequency deviations and tie-line power)
C = [1 0 0 0 0 0 0;    % delta_f1
     0 0 0 1 0 0 0;    % delta_f2
     0 0 0 0 0 0 1];   % delta_Ptie

% Feedthrough Matrix D
D = zeros(3,2);

% Create state-space model
sys = ss(A, B, C, D);

% Simulation time
t = 0:0.1:30;

% Input signal (step load disturbance in Area 1 only)
u = zeros(length(t), 2);
u(:,1) = deltaP_L1;     % Step load disturbance in Area 1
u(:,2) = deltaP_L2;     % No load disturbance in Area 2

% Simulate system response
[y, t, x] = lsim(sys, u, t);
```

```
% Plot Results
figure(1);
subplot(3,1,1);
plot(t, y(:,1), 'LineWidth', 2);
grid on;
title('Area 1 Frequency Deviation (Hz)');
xlabel('Time (sec)');
ylabel('\Delta f_1 (Hz)');

subplot(3,1,2);
plot(t, y(:,2), 'LineWidth', 2);
grid on;
title('Area 2 Frequency Deviation (Hz)');
xlabel('Time (sec)');
ylabel('\Delta f_2 (Hz)');

subplot(3,1,3);
plot(t, y(:,3), 'LineWidth', 2);
grid on;
title('Tie-Line Power Deviation (pu MW)');
xlabel('Time (sec)');
ylabel('\Delta P {tie} (pu MW)');
```